

spatial.ipynb - A simple script which runs coregistration, writes bem surfaces and bem solutions. This should be the first thing to run after preprocessing.

Requirements: MRI folder for participants (either anatomical or fsaverage)

Deepsources_Localization_using_beamformer.ipynb - Main pipeline for source localizing the cerebellum using a beamforming approach. The second half of the script creates an averaged source space for morphing participants to. I have used fsaverage for adults and vml_avg_child for children, but need to go back to using fsaverage for everyone in order to compare the activation locations in mni coordinates across groups consistently. Contact me (maxgarson@gmail.com or on slack) if you have questions about this. The last half of the script plots source time courses in subplot format for easy comparison.

Requirements: Epoched/Evoked data, outputs from **spatial.ipynb**, Empty room recording noise (erm_noise_cov) file.

Power_Analysis_Max.ipynb - Pipeline for spectral analysis. Creates a PSD plot for one epoch at a time, then averages them together across participants/conditions. Designed for RH cerebellum and LH motor, but the anatomical regions can be changed with some work. Normalizes the PSDs and computes the baseline PSDs as well (baseline from -500-0ms in the epochs). Runs 2-way ANOVA stats to find age, condition and interaction effects. Runs the FOOF package to extract summary features from the averaged PSDs: aperiodic slope, offset and intercept. See this publication for how we can use these metrics in our analysis/what we want to do:

<https://www.sciencedirect.com/science/article/pii/S1878929321000608?via%3Dihub>

Requirements: Epochs, forward solution, beamforming filters

Age_Analysis_VML_OFFICIAL.ipynb - Slightly outdated script, mostly for making figures for my BBND poster (the hard copy is in the lab if you want to look). First part of the script is cool: For each participant I plot the timecourses of activation in the ipsilateral cerebellum and contralateral motor regions, find the time points of the activation peaks and visualise these all together. Can be useful if we want to extract timing information or compare task dynamics across participants.

Requirements: Epochs, forward solution, beamformer filters

Paper_Figures- Plays with PSD data generated in Power_Analysis_Max and does some more statistics.

LCMV_DS.ipynb - ignore this outdated code

VML_DEEPSOURCE_STC.ipynb - ignore this outdated code

Timecourse_Smoothing_and_Peak_Extraction.ipynb - Simple script I wrote for Jieun that reads in STC files, extracts time courses from a set of labels, smooths them, and plots the TCs as an output.